Wei Zhang | CV

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Work and Educational Background

Karolinska Institutet–Department of Clinical Science(Postdoctoral Researcher) 2025.06 –

Harbin Institute of Technology–Faculty of Computing (Assistant Researcher) 2025.01 –

Harbin Institute of Technology– School of Mechanical and Electrical Engineering –Mechanical Engineering (PhD) 2019.09 – 2024.12

- Swarm Intelligence and Human-Computer Interaction Research Lab | Supervisors: Professor Zhao Tiejun
- Graduation Thesis: Research on Low-Dose CT Image Denoising Based on Deep Learning

Royal Institute of Technology–Division of Neuronic Engineering (Joint PhD Program) 2024.03 – 2025.03

- Laboratory: Department of Biomedical Engineering and Health Systems | External Supervisor: Prof. Xiaogai Li
- Funding: Harbin Institute of Technology Top Project

Harbin Institute of Technology - Faculty of Computing - Language Technology Center (Research Assistant) 2019.03 – 2019.09

Harbin Engineering University (HEU) – School of Naval Architecture and Ocean Engineering – Robotics(Master) 2016.09 – 2019.03

- Key Laboratory of Underwater Robotics | Supervisors: Professor Ye Li
- Graduation Thesis: Surface target detection based on unmanned surface vessels

HEU – School of Naval Architecture and Ocean Engineering –Structure Design (Undergraduate) 2012.09 – 2016.06

Area for Research

Medical image processing, Deep learning algorithms, Robot vision, Remote sensing image processing

Internship experience

Harbin Institute of Technology Robotics Group Tianyu Rehabilitation Medical Robotics Co., Ltd. Research Intern 2021.12 –2022.3

Pengcheng National Laboratory – Artificial Intelligence Research Center 2020.8 – 2021.8

Key highlights

1 First-author or corresponding-author works on low-dose CT denoising and PET/CT related topics, including recent conference papers on PET/CT fusion and image quality assessment.

2 Experience across academia and applied research, including clinical imaging collaboration and model validation tasks.

3 Granted invention patents related to multimodal sensing and intelligent perception.

4 Hands-on skills in deep learning model development, evaluation, and reproducible research pipelines.

List of papers and patents

Published Papers

[1] Zhang Kaining, Wang Jianfeng, **Wei Zhang***. MQ-UNet: Boundary-Aware Adapter Tuning of SAM2 for Colorectal Polyp Segmentation[C]. 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).

[2] Xiangyu Guo, Xiyuan Lu, Mingqiang Wei, Jianfeng Wang, **Wei Zhang***. Frequency-enhanced and conflict-adaptive ODE framework for consistent video editing without training[C]. 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).

[3] Yunzhe Huang, Jun Liu, Junjie Huang, Jinpeng Guo, Yanhong Zhang, Yuzhu Wu, Zemin Li, Wenxue Cui, Dezhen Wang, **Wei Zhang***. Efficient Modeling of Indoor Electromagnetic Signal Propagation in NLOS Environments Using a Gated MLP[C]. 2026 Asia-Pacific International Symposium on Electromagnetic Compatibility(APEMC).

[4] **Wei Zhang**, Jun Liu, Dezhen Wang, Zheng Cong, Jianfeng Wang, Wenxue Cui. Electromagnetic Environment Aware Quality Assessment for CT Images via Structured Artifact[C]. 2026 Asia-Pacific International Symposium on Electromagnetic Compatibility(APEMC).

[5] Wenhao Zhao, Cewen Tian, Xingyu Wang, Dezhen Wang, Xiaoyun Ye, **Wei Zhang***. MEU-Miner: Bitset-Guided Multi-Dimensional High-Utility Sequential Pattern Mining with Parallelism and Adaptive Top-k[J]. Knowledge-Based Systems, 2026.

[6] Yu Haoyang, Wang Jianfeng, **Wei Zhang***. FRAMU: A Lightweight Fractal Framework with Abductive Reasoning for Skin Leision Segmentation[J]. ACM Transactions on Multimedia Computing Communications and Applications, 2026.

[7] **Wei Zhang**, Xu Zi, Wang Jianfeng. GBA-UNet: An Ultra-Lightweight Region rowing Boundary-Aware UNet for Skin Lesion segmentation [C]. 2026 national Joint Conference on Neural Networks (IJCNN).

[8] Gao, R., Hu, Y., Zheng, D., Shao, M., Zhu, H., Song, C., ... & **Wei Zhang**. A Ranking Scheme for Trust Region Multi-agent Reinforcement Learning[C]. 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).

[9] Yuzhu Wu, Junjie Huang, Yizhe Wang, Yujian Bao, Zishen Liu, Xiaoyan Cao, Zheng Cong, Hongyang Wang, and **Wei Zhang***. YOLO-DCS: A Brain Tumor Detection Algorithm Based on Dynamic Self-Attention Mechanism and Adaptive Fusion[C]. 2025 International Conference on Bioinformatics and Biomedicine (BIBM).
[10] Wang Hongyang, Liu Tonghua, Song Chenyue, Li Siqiao, Yang Ming, Cong Zheng, **Wei Zhang***, and Li Fangwei. Denoising Low-dose Liver CT Using Generative Adversarial Networks with Perceptual Loss[C]. 2025 International Conference on Bioinformatics and Biomedicine (BIBM).
[11] **Wei Zhang**, Wenhao Jiang, Chenyue Song, Siqiao Li, Yongying Tan, Yuzhou Zhu. CMC-GAN: Cross-Modal Coupled GAN with Double Discriminators for PET/CT Image Fusion[C]. 2025 International Conference on Bioinformatics and Biomedicine (BIBM).
[12] **Wei Zhang**, Wenhao Jiang, Yongying Tan, Yuzhou Zhu. Structure-Aware Medical lmage steganography with ROl Protection and Reinforcement Learning Optimization[C]. 2025 International Conference on Bioinformatics and Biomedicine (BIBM).
[13] Zhao X, **Wei Zhang***. Improving yield by breaking the seed furrow and covering the soil after sowing in strip-tillage mode[J]. Plant, Soil and Environment, 2025.
[14] Chenyue Song, Chen Hui, Qing Lin, **Wei Zhang**, Siqiao Li, Haiqi Zhu, Zhixuan Li, Shengping Zhang, Shaohui Liu, Feng Jiang, Xiang Li. LVPNet: A Latent variable based Prediction driven End to end Framework for Lossless Compression of Medical Images[C]. 2025 International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI).
[15] Siqiao Li, Chen Hui, **Wei Zhang**, Feng Jiang, et al. MS-IQA: A Multi Scale Feature Fusion Network for PET/CT Image Quality Assessment[C]. 2025 International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI).
[16] **Wei Zhang**, Luo Yayuan, Fine-grained control over AI image generation: A comparative study of structured and unstructured cue words and their lightweight fine-tuning[J], CAAI Transactions on Intelligent Systems, 2025.
[17] Chenyue Song, Chen Hui, **Wei Zhang**, Shaohui Liu, Haiqi Zhu, Feng Jiang. BPCLIP: A Bottom up Image Quality Assessment from Distortion to Semantics Based on CLIP[C]. 2025 IEEE International Conference on Multimedia and Expo (ICME).
[18] **Wei Zhang**, Salmi A, Yang C, et al. Innovative Noise Extraction and Denoising in Low-Dose CT Using a Supervised Deep Learning Framework[J]. Electronics, 2024.
[19] **Wei Zhang**, Salmi A, Jiang F, et al. Enhancing Pulmonary Nodule Detection Rate using 3D Convolutional Neural Networks with Optical Flow Frame Insertion Technique[J]. IEEE Access, 2024.
[20] Salmi A[#], **Wei Zhang** [#], Jiang F. Reducing Data Transmission Efficiency in Wireless Capsule Endoscopy through DL-CEndo Framework: Reconstructing Lossy Low-Resolution Luma Images and Improving Summarization[J]. Mobile Networks and Applications, 2024.
[21] **Wei Zhang**, Jin W, Rho S, et al. A federated learning framework for brain tumor segmentation without sharing patient data[J]. International Journal of Imaging Systems and Technology, 2024.
[22] Song C[#], **Wei Zhang**[#], Jiang F, et al. ADTAH: Neuron 3D Reconstruction Via Adaptive Distance Transformation and Adaptive Hessian Matrix[C] 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). (EI)
[23] Shao, M., Zhu, H., Zhao, D., Han, K.., & **Wei Zhang**. Progressively Learning to Reach Remote Goals by Continuously Updating Boundary Goals. IEEE Transactions on Neural Networks and Learning Systems, 2024.
[24] Zheng D, Zhu H, Gao R.., & **Wei Zhang**. MMR-Sleep: A Multi-Channel and Multi-Receptive Field Sleep Stage Recognition Model[C]. 2024 Chinese Conference on Pattern Recognition and Computer Vision (PRCV).
[25] **Wei Zhang** Gao X, Yang C, et al. A object detection and tracking method for security in intelligence of unmanned surface vehicles[J]. Journal of Ambient Intelligence and Humanized Computing, 2022.
[26] **Wei Zhang**, Jiang F, Yang C F, et al. Research on unmanned surface vehicles environment perception based on the fusion of vision and lidar[J]. IEEE Access, 2021.
[27] **Wei Zhang**, Cui W, Jiang F, et al. Protecting the Ownership of Deep Learning Models with An End-to-End Watermarking Framework[C]. IEEE 20th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), 2021.
[28] **Wei Zhang**, Yang C, Jiang F, et al. Safety helmet wearing detection based on image processing and deep learning[C] 2020 International Conference on Communications, Information System and Computer Engineering (CISCE), 2020.
[29] **Wei Zhang**, Yang C, Jiang F, et al. A water surface moving target detection based on information fusion using deep learning[C] Journal of Physics: Conference Series. IOP Publishing, 2020.
[30] **Wei Zhang**, Yang C, Jiang F, et al. A review of research on light visual perception of unmanned surface vehicles[C] Journal of Physics: Conference Series. IOP Publishing, 2020.
[31] **Wei Zhang**, Yang C, Jiang F, et al. A super-resolution method of combined color image with depth map based on deep learning[C] Proceedings of the 2020 International Conference on Cyberspace Innovation of Advanced Technologies, 2020.
[32] **Wei Zhang**, Liao Yulei, Jiang Feng, et al. Review and trend analysis of unmanned surface vehicle technology development[J]. Unmanned Systems Technology, 2019.

Preprint Paper

[1] Chenyue Song, Chen Hui, Haiqi Zhu, Feng Jiang, Yachun Mi, **Wei Zhang**, Shaohui Liu. gmenting and Understanding: Region-aware Semantic Attention for Fine-grained Image Quality Assessment with Large Language Models.

[2] Meng Wan, Tangwu Zhong, Wenyang Shi, Jianyu Shen, **Wei Zhang** and Yizhe Sun. DM1 Drives Chemoradiotherapy-associated Enrichment of Adaptive NK Cells in Cervical Cancer.

Uder Rview

[1] Nie Rong, Wang Jianfeng, Li Xinag, **Wei Zhang***. A hierarchical modal gating and confidence fusion method for transaction graph fraud detection [J]. *CAAI Transactions on Intelligent Systems*, (EI)

[2] **Wei Zhang**, Wang Jianfeng. Substructure and Gene Group Aware Hierarchical Network for Explainable Dose-Response Curve Prediction[J]. *Nature Computational Science*, (SCI)

[3] Wang Siqi, Wang Jianfeng, **Wei Zhang***. CLEO: A Closed-Loop Framework for Synthesizing Medical Privacy-Preserving Tabular Data [J]. *npj Digital Medicine*, (SCI)

[4] **Wei Zhang**, Huang Junjie, Li Fangwei, Wang Jianfeng. Dynamic Weighted Fusion of MLP and TabPFN for Predicting Rupture Risk of Intracranial Aneurysms: A Multicenter Study [J]. *Radiology*, (SCI)

[5] Tan Yongying, Zhu Yuzhou, Feng Jiang, **Wei Zhang***. A review of low-dose CT image denoising studies [J]. *Medicine Advances*, (EI)

[6] **Wei Zhang**, Xu Zi, Wang Jianfeng. GBA-UNet: An Ultra-Lightweight Region rowing Boundary-Aware UNet for Skin Lesion segmentation [C]. *2026 national Joint Conference on Neural Networks* (IJCNN), (EI)

[7] **Wei Zhang**, Wenhao Jiang, Jianfeng Wang, Xiaochun Cheng, Junjie Huang, Yuzhu Wu, Hongyang Wang, Mingqiang Wei. Task Aware Learned Medical Image Compression via Feature Total Variation Regularization and Joint Optimization [C]. *35th International Joint Conference on Artificial Intelligence*(IJCAI)

[8] Jianfeng Wang, **Wei Zhang**, Xiaochun Cheng, Yu Geng, Mingqiang Wei. Training Free Emotion Recognition in Multimodal Large Language Models via Confusion Guided Attention and Confidence Adaptive Pruning[C]. *35th International Joint Conference on Artificial Intelligence*(IJCAI)

Authorized Invention Patent

[1] Wang Bo; **Wei Zhang**; Jiang Kunyi; Liao Yulei; Li Ye; Feng Jiaxiang; Su Guangsheng; Zhou Bin; Yuan Zhihao; Yang Shiyuan; Zhang Lei; Su Yumin; Jia Zhihao. "A method for surface target detection, identification and localization of unmanned surface vessels based on monocular camera and lidar information fusion." CN109444911B, 2023.

[2] **Wei Zhang**; Liao Yulei; Li Ye; Wang Bo; Chen Qixian; Jia Zhihao; Wang Youkang; Pang Shuo; Zhuang Jiayuan; Zhang Lei; Su Yumin. "An improved method for online speed prediction of unmanned surface vessels based on BP neural network." CN108960421B, 2022.

[3] **Wei Zhang**; Li Ye; Wang Bo; Liao Yulei; Chen Qixian; Jia Zhihao; Wang Youkang; Pang Shuo; Jiang Quanquan; Gu Huayu; Wu Haowei. "A method for online detection and identification of deep-sea targets based on multi-threshold dynamic statistics." CN109344717B, 2021.

[4] **Wei Zhang**; Li Ye; Wang Youkang; Liao Yulei; Li Yiming; Wang Leifeng; Pan Kaiwen; Zhang Weixin. "A method for predicting the actual heading of a wave glider." CN107942095B, 2019.

[5] **Wei Zhang**; Li Ye; Wang Youkang; Liao Yulei; Li Yiming; Wang Leifeng; Pan Kaiwen; Zhang Weixin. "A method for autonomous berthing and departure of unmanned surface vessels in a pool using multi-sensor information fusion." CN109213162A, 2019.

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[1] **Wei Zhang**, Li Ye, Wang Youkang, et al. Target recognition in complex environments using feature matching.

[2] **Wei Zhang**, Li Ye, Wang Youkang, et al. Judging various image filtering and edge sharpening fusion optimization procedures.

Research Project Experience

[1] **Research on Key Technologies for Denoising and Quality Assessment of Low-Dose CT Images, Provincial Natural Science Foundation, 2020.01–2023.12, Key Participant**

Participated in the design of algorithms for denoising and quality assessment of low-dose CT images, proposing a multi-scale deep learning denoising model and image quality prediction framework. This significantly reduces radiation dose while preserving diagnostic information. The algorithm has been applied to PET/CT image quality assessment experiments and provides a technical foundation for subsequent clinical validation. Two SCI papers have been published related to this work.

[2] **Research on Algorithm for Predicting the Risk of Cerebral Aneurysm Rupture, Provincial Natural Science Foundation, 2020.01–2023.12, Key Participant**

First proposed a cerebral aneurysm rupture risk prediction model based on hemodynamic characteristics and deep learning, achieving automatic risk assessment and individualized prediction. Combined with interpretability methods such as SHAP, the correlation between key blood flow parameters and aneurysm rupture risk was revealed, providing interpretive evidence for clinical decision-making. The relevant work has been submitted to Nature Cancer.

[3] **Research on Lung Cancer Survival Prediction and Treatment Recommendation Algorithms, Pengcheng National Laboratory Fund, 2021.01 – 2024.12, Key Participant**

Participated in multimodal data modeling of lung cancer patients, including the integration of pathological sections, images, and clinical indicators to achieve survival prediction and treatment plan recommendation. Proposed a data

augmentation strategy based on generative models to effectively solve the small sample size problem and improve the model's generalization ability.

[4] **Research on Multimodal Fusion and Quality Assessment Methods for Medical Imaging, Horizontal and Vertical Integrated Project, 2019.12 – 2022.12, Key Participant**

Researched PET/CT multimodal fusion technology, proposed a GAN-based image reconstruction method to improve the joint representation quality of metabolic and anatomical information. Developed a comprehensive analytical framework for fusing image quality assessment and clinical indicators for disease grading and prognostic prediction.

[5] **Key Technologies and Core Systems for Intelligent Analysis and Recommendation Distribution of Multimedia Content Quality, Ministry of Science and Technology, Key Research and Development, November 2020 – October 2023, Participant**

I participated in the design of the quality assessment algorithm in this project and proposed an image quality assessment method based on compressed sampling. This method can achieve results comparable to the optimal algorithm with only 30% of the original information, and improves performance by about 10%. The related work has been submitted to the CCF-B category conference ICASSP.

[6] **Research on Key Technologies for Early Diagnosis and Prognostic Prediction of Infant Brain Injury, Call of Life Foundation, Royal Institute of Technology, Sweden, April 2024 – December 2024, Principal Investigator**

Focusing on early diagnosis and neurological function prediction of hypoxic-ischemic brain injury (HIE) in infants, this project combines multimodal imaging such as MRI and DWI with clinical indicators for intelligent analysis. An image analysis framework based on multi-scale feature fusion and interpretable deep learning was designed to achieve automatic lesion area detection, damage severity grading, and prognostic risk prediction. Explore the construction of an image-clinical knowledge graph and cross-modal causal analysis to reveal the injury mechanism and provide decision support for personalized treatment strategies.

Leading research project

Robust PET-CT quality assessment methods for clinical applications — Stiftelsen Lars Hiertas Minne

Academic Services

Adjunct Professor

- Visiting Master's Supervisor at Mudanjiang Medical University

Editorial Board Member

- International Journal of Medical Imaging (IJMI) –Associate Editor
- Journal of Clinical Diagnosis and Treatment –Editorial Board
- Asia Pacific Journal of Clinical Medical Research –Editorial Board
- Intelligence & Robotics –Editorial Board
- Journal of Intelligent Systems – Youth Editorial Board (Journal of the Chinese Association for Artificial Intelligence)
- Medicine Advances– Youth Editorial Board
- Biomedical Engineering Materials– Youth Editorial Board
- Interdisciplinary MEDICINE– Youth Editorial Board
- Medical Data Mining– Youth Editorial Board
- Intelligent Medical Science – Youth Editorial Board
- Journal of Hainan University – Youth Editorial Board

Journal Guest Editors and Conference Services

- Optics– Advancements in Optical Imaging and Sensing for Biomedical and Environmental Applications Guest Editor
- Intelligence & Robotics– Intelligent Medical Imaging and Medical Robotics: Al-Enabled Methods, Clinical Translation, and Responsible Deployment– Guest Editor
- Medicine Advances – Trustworthy Medical AI for Screening, Diagnosis and Prognosis: From Image Quality to Clinical Decision Support– Guest Editor
- Intelligent Medical Science–Quality Assurance and Data-Efficient Intelligent Healthcare– Guest Editor
- 24th International Conference on Cyberworlds (CW 2025) – Poster Chair
- IEEE SysCon 2026– Technical Program Committee
- 2026 Practice and Experience in Advanced Research Computing– Technical Program Committee

Reviewer

- IEEE Transactions on Neural Networks and Learning Systems（IEEE TNNLS）
- IEEE Transactions on Circuits and Systems for Video Technology（IEEE TCSVT）
- ACM International Conference on Multimedia（ACM MM）
- International Conference on Computer Vision 2025 （ICCV）
- CAAI Transactions on Intelligence Technology（CAAI TRIT）

- IEEE International Conference on Acoustics, Speech and Signal Processing （ICASSP）
- International Joint Conference on Neural Networks （IJCNN）
- IEEE International Conference on Multimedia Expo（ICME）
- Journal of Signal Processing Systems
- IEEE Conference on Computer Vision and Pattern Recognition（CVPR）
- IEEE Transactions on Medical Imaging
- IEEE Transactions on Multimedia

Award record

- Second Prize of Feasibility of OI China Underwater Robot Competition
- Innovative third prize in OI China Underwater Robot Competition
- Effective Return on Investment in The third Maritime RobotX Challenge, USA
- Successful particpation award in The third Maritime RobotX Challenge, USA
- 2018 Ministry of Industry and Information Technology Innovation and Entrepreneurship Second Class Scholarship
- The master's thesis is included in the full-text database of China's excellent master's thesis
- Advanced Individual of Harbin Engineering University's 60th Anniversary Celebration
- Outstanding Volunteer for the 60th Anniversary Celebration of Harbin Engineering University
- The third prize of the 3rd Campus DV Competition of Harbin Engineering University
- Outstanding Volunteers of the 2nd National Marine Vehicle Design and Production Competition
- Harbin Engineering University "Qihang Cup" College Student Science and Technology Innovation Competition Excellence Award
- Second Prize of Harbin Engineering University Student Science and Technology Innovation Competition
- Harbin Engineering University's scientific research projects are supported by ordinary type three times
- Harbin Engineering University "Qihang Cup" University Student Science and Technology Innovation Popularization Competition Excellence Award
- Harbin Engineering University 2013 Autumn Semester First Prize Scholarship
- Harbin Engineering University MARIC Inspirational Scholarship
- Harbin Engineering University 2015 Spring Semester First Prize Scholarship
- Excellent Communist Youth League cadres excellent youth Communist

Supervising, mentoring activities

- Video Frame Interpolation Technology Based on Deep Learning (Gu Donghao, 2020)
- Research on Interpretable Biological Image Analysis Algorithms (Li Yaowei, 2020)
- Research on pulmonary nodule detection technology based on deep learning (Wen Zhaojing 2020)
- Study on the efficacy of low-dose dual-source CT combined with Cycle GAN in displaying liver lesions (Weiqiang Zhao, 2020)
- Research on low-dose liver and lung CT denoising algorithm (Cui Yang, 2021)
- Study on the performance of low-dose dual-source CT combined with Cycle GAN in displaying renal lesions (Qi Zhang, 2021)
- Study on the three-dimensional morphological reconstruction of mouse neurons at the whole brain scale(Huang Jiafei,2023)
- Research on low-dose CT denoising based on multi-scale information (Li Yuzhou, 2023)
- Research on low-dose CT image denoising based on edge information(Tan Yongying, 2023)
- Survival prediction of cancer patients based on multimodal information(Chen yue, 2023)
- Study on 3D tracking and reconstruction of mouse neurons at the whole brain scale(Li kai, 2023)
- Method for correcting incorrect terminal points in neuronal 3D reconstruction structure(Zheng Deqin, 2023)